

Precision Bipolar Bias-Coil Current Driver — Revision 2

Design process, simulation verification, PCB layout and assembly

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Abstract

This document records how Revision 2 of the linear bipolar current driver for the magneto-optical-trap bias coils was designed and verified. The driver delivers up to ± 7 A into a series Helmholtz pair under closed-loop control of an SRS SIM960 analog PID controller, using a Danisense DP50IP-B fluxgate transducer as the feedback sensor. Revision 2 corrects the problems found on the first board (1V1), replaces the resistive pull-down that limited output headroom with a transistor current sink, adds supply protection and status indication, and moves to a 4-layer board with a clamped heatsink sandwich. Every design value was checked in a netlist-driven ngspice model in which each component carries a datasheet-validated model, including the bench supplies, the controller and the transducer. The PCB was checked against the schematic and simulated again with parasitics extracted from the actual copper.

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1 Purpose and requirements

The apparatus drives the magnetic bias coils of a cold-atom magneto-optical-trap experiment: three axes (X, Y, Z) for bias and shim fields and a fourth for the MOT gradient coils. Each axis needs bidirectional current through zero without a polarity-switching discontinuity, because the cancellation coils routinely operate near zero amperes. Four Rev 2 boards will be built, one per axis, and mounted in a 19-inch rack.

Item	Requirement / value
Load (per axis)	Series Helmholtz pair: two coils of $886\,\mu\text{H}$ / $0.656\,\Omega$ (measured $\tau = 1.35\,\text{ms}$), coupling $k \approx 0.01 \Rightarrow 1.79\,\text{mH}$, $1.312\,\Omega$
Output current	Bipolar, $\pm 7\,\text{A}$ design point (Rev 1 targeted $\pm 10\,\text{A}$; see Section 6.2)
Field stability	Field fluctuation below $1\,\text{mG}$ at the atoms (current equivalent: Section 1.1)
Current stability target	10 ppm, i.e. $60\,\mu\text{A}$ at $6\,\text{A}$, used as the design budget so far
Controller	SRS SIM960 analog PID (100 kHz bandwidth), setpoint from an NI DAQ analog output
Feedback sensor	Danisense DP50IP-B fluxgate, 4 primary turns (1:250, $12.5\,\text{A}$ nominal), $100\,\Omega$ burden $\Rightarrow 0.400\,\text{V/A}$
Supplies	Delta Elektronika ES015-10 ($15\,\text{V}$ / $10\,\text{A}$, floating outputs). One series pair per board for the power rails; one series pair shared by the four boards for the signal rails
Operating mode	Constant-voltage $15.0\,\text{V}$, current limit at maximum ($10\,\text{A}$)
Environment	Lab ambient 24°C ; continuous (worst-case) coil duty assumed

Table 1: Requirements and operating conditions.

Speed limit. Current slew into the coil is set by the forcing voltage, not the amplifier: $\Delta t = L \Delta i / V$. On $\pm 15\,\text{V}$ rails a full reversal takes several milliseconds; raising loop bandwidth speeds small steps but cannot accelerate large ones.

1.1 Field-noise requirement in current terms

The requirement is stated as a field: fluctuations below $1\,\text{mG}$. The current equivalent is $\delta I = \delta B / k_{\text{coil}}$, where k_{coil} is the coil pair’s field constant at the atoms (G/A). It has not been measured yet; the Rev 1 document used a “generous” $10\,\text{G/A}$ as an estimate. For a Helmholtz pair of N turns per coil and radius R , $k = (4/5)^{3/2} \mu_0 N / R$.

k_{coil}	1 mG equals	ppm of 7 A	DP50IP noise, same units
2 G/A	500 μA	71 ppm	
5 G/A	200 μA	29 ppm	0–100 Hz: 50 μA pk-pk (4 ppm of 12.5 A)
10 G/A	100 μA	14 ppm	0–1 kHz: 100 μA pk-pk (8 ppm), 87 μA rms

Table 2: The 1 mG requirement as a current, for plausible coil constants, against the DP50IP’s specified noise (referred to the primary).

Inside the loop bandwidth the coil current follows the sensor, so the DP50IP’s own noise reaches the coil up to the closed-loop bandwidth (≈ 2.6 kHz at $P = 30$). If k_{coil} is near 10 G/A, the sensor’s 0–1 kHz noise alone is comparable to the 1 mG budget, and the loop bandwidth should be reduced to what the experiment’s settling needs. A lower limit applies: the SIM960 error input saturates beyond about ± 1 V (± 2.5 A at 0.4 V/A), so $P \geq 10$ is needed to reach full output on a large step; $P = 10$ puts the crossover near 0.5 kHz.

TO SUPPLY: Coil field constant k_{coil} in G/A at the atom position (turns and radius, or a Hall/fluxgate gauss-meter reading at a known current), and whether “1 mG” means peak-to-peak or rms and over what bandwidth. These two numbers turn the field requirement into a current specification and decide the SIM960 gains.

2 Background: the legacy prototype and the first board (1V1)

2.1 Legacy prototype

Before the present architecture, a four-axis prototype used a series MOSFET in its linear region as a voltage-controlled resistor from a 30 V supply, an IGBT as a static switch, a LEM transducer (1:250, 100 Ω burden, 0.4 V/A) and an SRS LTS-series PID driving the MOSFET gate unbuffered. Two failures are recorded: the X-axis EVAL-ADuM4221 gate driver failed short (bypassed with a jumper), and a Y-axis heating incident (3.3 k Ω stand-in load, 770 ms square wave, smoke after 20 s) whose root cause was never established. Its grounding ties made floating supplies mandatory, which carried forward into the present design as floating ES015 supplies and a single deliberate SGND–PGND bond.

2.2 Board 1V1: four faults on first power-up

The 1V1 board (OPA455 driving a two-stage complementary emitter follower with five MJL21194G / MJL21193G pairs) did not work on first power-up. Four independent faults were found and fixed; clearing any three left the board non-functional.

Fault	Root cause	Fix on 1V1 (and Rev 2)
Op-amp output disabled	E/D pin unconnected; measured 0.06 V above E/D Com, inside the disable window	10 k / 4.7 k divider, 4.8 V (R33/R34)
No reference to the power section	Net tie NT1 had no footprint; SGND and PGND measured 200 k Ω apart	External bridge at the supply (R27 net tie on the board)
Oscillation near 3 MHz	Drivers Q13/Q14 driving the bank’s reflected input capacitance with no base stoppers	220 Ω stoppers via trace cuts (R35/R36)
Bias untrimmable	V_{BE} -multiplier floor: $R_2 I_B$ held 0.5–0.8 V; the best reachable bias was 875 mA/bank (26 W idle)	15.6 k across R2 ($R_{2,\text{eff}} = 6.09$ k; R2 = 6.04 k)

Table 3: 1V1 bring-up faults. The Rev 1 log quotes $R_{2,\text{eff}} \approx 5.86$ k; $10 \text{ k} \parallel 15.6 \text{ k} = 6.09$ k.

The oscillation. Measured with C22 lifted and no stoppers: 3.09 MHz (324 ns period), HV node ≈ 1.53 V pk-pk, NPN drive bus 1.28 V pk-pk, supplies at their 50 mA limits. With C22 fitted a second, C22-closed mode appeared at 1.67 MHz with a ≈ 3.7 MHz ring; it vanished when C22 was lifted and was never reproduced in simulation (attributed to supply current-limit interaction). A SPICE model built from the netlist, vendor models and PCB trace parasitics predicted 3.20 MHz / 1.58 V pk-pk (4 % and 3 % from the bench) and 3.73 MHz for the ring, and correctly predicted that stoppers of at least 200 Ω stabilise the full command range. After the fix a 3.70 MHz ring with 30 % overshoot (three cycles, 1.1 μ s settling) remained, possibly pickup rather than real current.

After the fixes. No oscillation; bias trimmed to 10 mA cold, settling to 50 mA warm in 15 minutes; incremental gain within 2 % of prediction, in phase and flat from 10 Hz to 1 kHz (into a 113 Ω resistive load); a 2.1 V offset at the output (closed-loop operation removes it). A later trim of the same board (30 September) set about 150 mA per bank; the trimmer setting and heatsink temperature for that point were not recorded. The 1V1 trimmer, labelled 5 k Ω , measured 0.6 Ω to 3 k Ω end to end (25 turns, clockwise = maximum).

Design audit. A netlist audit produced eighteen findings (four confirmed on the bench as the bring-up faults). Those findings, the 1V1 headroom limit from the 820 Ω pull-down, and the debugging experience define Revision 2:

- 220 Ω driver stoppers designed in (R35/R36); C22 kept, with R29 in series.
- The 820 Ω pull-down, which starved the PNP half and pinned the drive node near 11.3 V, replaced by a transistor current sink (Q15/Q16).
- U2 given a gain of 1.33 (it was a follower), so the sink's headroom gain is usable.
- E/D biased, status flag and DP50IP status brought to LEDs, command input given a DC return.
- Net tie with a real footprint; Zobel and output isolation added; test-point access considered.

Not yet measured on 1V1. Every bench measurement was open loop with the SIM960 in manual mode and below 1 A. No closed-loop noise or drift data exist yet. The open-loop record contains instrument floors (shorted probe 606 μ V rms; $\times 1$ probe with 20 MHz limit 275 μ V rms; 16.8 mV pk-pk on the monitor with the power stage off), the 16.7 kHz fluxgate excitation (confirmed by unpowering the transducer), and a ≈ 97 ms burst envelope from a switch-mode supply. One output NPN was found not conducting, suspected cold solder joint, unconfirmed.

TO SUPPLY: (1) RV2 on 1V1 at the 150 mA/bank trim: measure pin 1 to pin 2 with the board unpowered (the in-circuit reading is the pot), and note the heatsink temperature when it was set. (2) The suspect NPN: three unpowered readings (drive bus to base 2.2 Ω , emitter to COIL_DRIVE 0.22 Ω , collector to Power+ ≈ 0 Ω); one will be open.

3 Architecture

Supply and ground domains. Two independent supply pairs feed the board: *Power* $\pm$ (J11, output stage only) and *Signal* $\pm$ (J12, U2, the bias network and sink, the DP50IP and the indicators). Their grounds, PGND and SGND, are separate planes on layer 2 and join at exactly one point, the net tie R27. Because the supplies are floating, R27 is the only DC path between the domains, so no PGND current can flow in SGND regardless of where the tie sits. This lets U2 drive the output toward the rail while *Power* $\pm$ sags under load, and keeps coil-return current out of the precision ground.

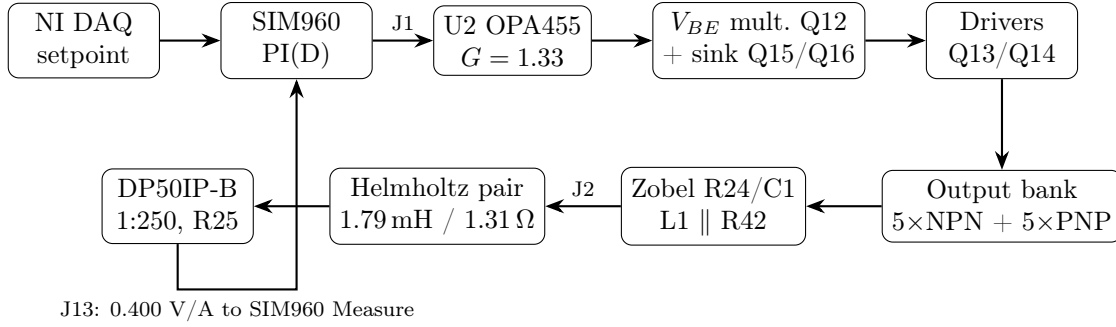


Figure 1: Signal path. The coil return passes four times through the DP50IP primary before reaching PGND.

4 Circuit design

4.1 Command input and gain stage

J1 receives the SIM960 output through R28 (100 Ω) with C21 (100 pF) to SGND (corner 15.9 MHz). R32 (47k) returns U2’s input when J1 is unplugged, and R43 (680k) from Signal+ biases the command so that an unplugged J1 gives -28 mA in the coil rather than -0.9 A . U2 (OPA455, HSOIC PowerPAD on Signal-) is a non-inverting stage with $G = 1 + 3.3\text{ k}/10\text{ k} = 1.33$. Its E/D pin is biased to 4.80 V by R33/R34 (enable threshold 0.8–5.5 V above E/D Com; absolute maximum 7 V). Its open-drain status flag drives D12 (red, “U2 FAULT”) from Signal+ through R48; the flag is referenced to E/D Com and may be pulled to V+.

4.2 Bias spreader and trim

Q12 (BD139-16, thermally coupled to the heatsink) forms a V_{BE} multiplier between the NPN and PNP predrive nodes, with R2 (6.04k, 0.1 %) on top and R23 + RV2 (rheostat) below. R26 (10 Ω) is the base stopper at Q12’s base pin; C22 (1 μF) + R29 (22 Ω) across the spreader damp the drive network. RV2 is wired so that a wiper failure goes to the *high-resistance, zero-bias* end (fail-safe).

A finding from the verification (Section 6.1) is that Q12’s base current through R2 contributes about 0.7 V of the spreader voltage, so the trim point depends on Q12’s h_{FE} . The values were re-chosen so that the full h_{FE} range (100–250) and temperature extremes are covered: **R23 = 1.5 k** (was 2.2 k), RV2 = 5 k, 25-turn.

4.3 Current sink

Q15/Q16 (BD139-16) with R37 and R38 (47 k) form a constant-current sink from the PNP predrive node to Signal-, with R39 (22 Ω) in Q15’s collector. It replaces the 1V1 pull-down. The sink current flows through U2’s output at high positive output, so it directly reduces U2’s output swing. **R37 was raised from 33 Ω to 56 Ω** (sink 18 $\rightarrow$ 10.8 mA), which raised the positive current limit from +7.31 A to +7.69 A (Section 6.2). Check value at bring-up: about 0.6 V across R37.

4.4 Drivers and drive network

Q13/Q14 (MJE15032G/MJE15033G) drive two base buses. R35/R36 (220 Ω) are the driver base stoppers that cured the 1V1 oscillation. R1 (47 Ω , 2512) bleeds between the buses. D3/D4 (B340A)

and D5–D7 (1N4148W) protect base–emitter junctions against reverse voltage (without D4/D7 the reverse V_{EB} reaches 5.9/7.4 V against a 5 V limit).

4.5 Output bank

Five MJL21194G and five MJL21193G (TO-264, horizontal, tab up under the heatsink), each with a $2.2\,\Omega$ base resistor and a $0.22\,\Omega$ emitter ballast (Isabellenhütte SMT-R220-1.0: 1 %, ± 50 ppm/K, < 3 nH, 7 W). Non-inductive ballasts and base resistors are mandatory: series inductance in the emitter defeats the degeneration at the frequencies where the follower input impedance turns negative. With matched-lot devices the simulated sharing spread was at most $1.13\times$.

4.6 Output network

A Zobel R24 ($4.7\,\Omega$, 2 W metal film, non-inductive) + C1 (220 nF, 100 V) at COIL_DRIVE, and an output isolator L1 ($2.2\,\mu\text{H}$ IHLP-5050FD) in parallel with R42 ($2.2\,\Omega$, 2 W) between COIL_DRIVE and the coil terminal. L1||R42 removed every cable/coil-capacitance resonance case found in the first verification round. D1/D2 (MUR1560G) catch the output to the rails.

4.7 Sense chain

The DP50IP-B is wired for four primary turns (pins 5→1→6→2→7→3→8→4), ratio 1:250. Its secondary current I_s (pin 9) flows through R25 ($100\,\Omega$ 0.1 %, 10 ppm/K, FRH2512) to SGND, giving 0.400 V/A at J13. Supply draw is 42 mA + I_s from +15 V and 28 mA + I_s from −15 V, about 80 mA at full current. The STATUS optocoupler (pins 13/14) conducts when the sensor is healthy and shunts D13 (red, “SENSOR FAULT”), which lights when the opto opens. J10 brings the status out.

4.8 Signal-supply protection and indicators

At J12 each signal rail has a PTC fuse (Bourns MF-MSMF030: hold 0.30 A, trip 0.60 A, 30 V, max 10 A), an SMBJ14A TVS and a B340A Schottky (reverse-polarity shunt). The rail LEDs D8–D11 (green, 6.8 k, about 1.9 mA) show power present and that the bulk capacitors are still charged. The complete schematic is reproduced in Appendix B.

5 Verification method

5.1 Netlist-driven simulation

A script converts the KiCad netlist (later, the PCB’s own pad connectivity) directly into an ngspice deck. Every component is emitted with a model; an unknown part stops the generator, so nothing is silently omitted. External elements are modelled as well: the coil pair and its cable, both ES015-10 supply pairs and their cables, three further boards on the shared signal supply, the SIM960, and the BNC cables. Operating points are found with a power-up transient (supplies ramped as the ES015 does) rather than a DC solve, which proved more robust with the behavioural sub-models.

5.2 Sub-models and their validation

Model	Basis	Check against datasheet
OPA455	Behavioural, from SBOSA16 (the supplied zip was an OPA462 TINA file)	Aol 135 dB; unity crossing 5.5 MHz (GBW 6.5 MHz); PM 57° (ds $\approx 54^\circ$); SR 31.3 V/ μ s (32); I_{SC} +43/−49 mA; swing 3.0/2.5 V at 20 mA; E/D and flag correct
MF-MSMF030 PTC	Electro-thermal: $R(T) = R_0 e^{(T-23)/30}$, trip above $\approx 112^\circ\text{C}$, R_{th} 128 K/W, C_{th} 0.053 J/K	No trip at 0.30 A/23 °C and at derated holds; 8 A trips in 0.039 s (max 0.10 s); tripped 0.65 W (0.8 W typ)
SMBJ14A	Diode, V_{BR} corners 15.6/16.4/17.2 V	V_C fitted to 23.2 V at 25.9 A
ES015-10	Source-only regulator, CC limit, 2000 μ F internal	Fall time 6.7/66 ms vs. 7/70 ms; load-step deviation 147 mV (max 160 mV)
DP50IP-B	1:250 current source, 600 kHz pole (conservative), supply currents per datasheet	—
SIM960	From the manual: $\varepsilon = \text{SP-M}$, input saturates $\approx \pm 1.17$ V, $P\varepsilon$ saturates ± 10.5 V, conditional integration, soft limiter, 100 kHz, 1 μ s delay, 1 M Ω Measure input	—
Diodes	Vendor SPICE (Diodes Inc., onsemi MURF1560)	—

5.3 Transistor models

Two model sets were carried through: the vendor files and a datasheet-fitted set. The vendor MJL21193/21194 models have a forward ideality $NF \approx 0.85$, which is non-physical; at operating currents the two sets differ by about 0.17 V in output-device V_{BE} (0.49–0.52 V vs. 0.68 V at 150 mA) and in BD139 V_{BE} (0.82 vs. 0.65 V at 2.5 mA). With the vendor set, 140 mA/bank was unreachable anywhere on the trimmer (0.39–3.3 A/bank even at RV2 = 5 k). Because 1V1, with the same device types and the same spreader network after the R2 fix, trimmed to 10–50 mA and later to about 150 mA per bank, the vendor set is contradicted by measurement and the fitted set is used as primary.

TO SUPPLY: Measured V_{BE} versus I_C on spare parts (BD139-16 at 0.3, 2.5, 10 mA; MJL21194 at 30, 150 mA, 1 A) would pin the absolute bias predictions and resolve the model question.

6 Verification results

6.1 Bias trim (final values)

Corners: Q12 h_{FE} 100/160/250; temperatures “max bias” (outputs and drivers 70 °C, Q12 25 °C), nominal (all 30 °C), “min bias” (outputs 24 °C, Q12 60 °C).

h_{FE}	RV2 = 5 k	140 mA/bank at	1 A/bank at	RV2 = 0
100	0 / 1.8 mA / 0	1563 / 2794 / 996 Ω	994 / 1777 / 577 Ω	5.4 / 7.8 / 3.8 A
160	0 / 0.06 mA / 0	1127 / 1967 / 716 Ω	703 / 1278 / 384 Ω	4.6 / 6.8 / 3.1 A
250	0 / 0.01 mA / 0	920 / 1608 / 576 Ω	558 / 1046 / 284 Ω	4.1 / 6.3 / 2.6 A

Table 5: Bias per bank versus RV2 lower-leg resistance ($R_{23} = 1.5\text{ k}$, $R_{37} = 56\text{ }\Omega$). Each cell lists nominal / max-bias / min-bias corners.

The deadband is reached at the 5 k end in every corner, 1 A is reachable in every corner, and the 140 mA trim lands at 12–56 % of travel. On a 25-turn 5 k trimmer one tenth of a turn changes the bias by about 9 %. Larger R_{23} values (1.8 k, 2.0 k) lose the 1 A reach. Idle bias falls slightly as the board warms (131–140 mA at a 29 °C heatsink), which is thermally stable.

Trimmer range. The 1V1 trimmer measured only $\approx 3\text{ k}\Omega$ end to end. If the new part also stops near 3 k Ω , the 140 mA point is still reachable in every corner (worst 2794 Ω), and the zero-bias end still works in the nominal and min-bias corners (conduction starts at 1.0–2.6 k Ω), but not in the extreme max-bias corner. Measure the Rev 2 trimmer end to end before fitting.

6.2 Output current limits

With the real supplies the positive limit is set by U2’s output swing from the 15 V signal rail while sourcing the sink current. Raising R_{37} trades sink current for swing:

R37	Sink	+limit, cold coil	+limit, coil +20 K	–limit, cold
33 Ω	18.2 mA	+7.31 A	+6.83 A	–9.21 A
47 Ω	12.8 mA	+7.58 A	+7.09 A	–9.18 A
56 Ω	10.8 mA	+7.69 A	+7.19 A	–9.16 A
68 Ω	8.9 mA	+7.78 A	+7.27 A	–9.14 A

Table 6: Current limits versus R_{37} and coil temperature.

The positive output reaches about 10.6 V at the coil terminals, so the positive ceiling falls as the copper warms (+3930 ppm/K):

Coil rise	R (pair)	+limit	–limit	Coil dissipation at 7 A
0 K	1.312 Ω	+7.70 A	–9.54 A	64 W
20 K	1.415 Ω	+7.20 A	–8.92 A	69 W
40 K	1.518 Ω	+6.77 A	–8.38 A	74 W
60 K	1.621 Ω	+6.38 A	–7.89 A	79 W
100 K	1.828 Ω	+5.73 A	–7.07 A	90 W

Table 7: Output ceilings against coil temperature rise.

+7 A is lost at about 29 K of coil rise, which requires a coil thermal resistance of at most $\approx 0.42\text{ K/W}$ at $\approx 69\text{ W}$, demanding for a bare coil in still air. The negative side keeps –7 A to about 80 K. The $\pm 7\text{ A}$ specification is therefore conditional on the measured coil temperature rise.

TO SUPPLY: Coil temperature rise at 7 A: measure the pair’s resistance cold, run at 7 A until the reading stops drifting (likely 20–40 minutes), measure hot; $\Delta T = (R_{\text{hot}}/R_{\text{cold}} - 1)/0.00393$. Also the coil pair’s self-resonant frequency (function generator through a known series resistor, sweep for the impedance peak); it matters only if it falls near the 154 kHz Zobel corner.

6.3 Open loop and closed loop

Test	Result
Open-loop DC transfer	$g_m = 0.9147 \text{ A/V}$, zero at 1.03–1.05 V, linearity residual 27 mA over $\pm 7 \text{ A}$, limits +7.69/–9.14 A, sense 0.39997 V/A
Open-loop frequency response	Flat 0.915 A/V to the coil pole ($\approx 129 \text{ Hz}$), smooth to 100 kHz
Loop gain, $P = 30$, $I = 2400 \text{ s}^{-1}$	Crossover 1.44–1.45 kHz, phase margin 79–80°, gain margin > 35 dB; closed-loop bandwidth $\approx 2.6 \text{ kHz}$, peaking 0.5 dB
Loop gain, $P = 120$	Crossover 5.1 kHz, phase margin 86° (ample gain headroom)
Setpoint steps 10 mA / 0.5 A / 1 A / $\pm 7 \text{ A}$	Overshoot $\leq 8.4 \%$, settle to 0.1 % in 1.9–4.0 ms
Reversal +7 $\rightarrow$ –7 A	Settles in 3.8–4.6 ms; rails rise only to –15.6 V (supplies cannot sink); peak 27 W per output device, 2.4 W per driver
Triangle $\pm 7 \text{ A}$ at 1 / 10 Hz	Tracking error 1.1 / 11 mA rms (PI ramp error, $K_v \approx 2.6 \times 10^4 \text{ s}^{-1}$)
Triangle $\pm 7 \text{ A}$ at 100 Hz	Not achievable: needs $\approx 15 \text{ V}$ across the coil vs. $\approx 10.5 \text{ V}$ available
DC scale	+70 ppm from the SIM960’s 1 M Ω Measure input loading R25 (fixed; calibrate out)

Table 8: Signal-path verification (fitted models, ES015 supplies, four boards on the signal pair).

6.4 Stability against parasitics

Single-parameter extremes (predrive trace to 600 nH, Q12 traces to 400 nH, lead and bus inductance, output-pour capacitance 0–3 nF, coil capacitance 10 pF–10 nF, cables, U2 output impedance 50–120 Ω , U2 bandwidth $\pm 30 \%$, MLCC DC-bias derating, coil disconnected) and the all-worst combination showed no oscillation. An impulse test (0.5 A, 50 ns into the output; 2 mA into the predrive) with parasitics extracted from the routed board decays to 2–3 % within 12–18 μs ; with three times those parasitics plus 10 nF coil capacitance and worst-case U2 it decays to 8–12 %.

6.5 Component stress and supply currents

At idle, $\pm 7.5 \text{ A}$ and both limits every component is below 52 % of its rating, except the SMBJ14A, which sits at 14.95 V against its 14 V stand-off (inherent in clamping below the DP50IP’s 16.5 V maximum; keep the signal supply at or below 15.0 V). Signal-rail draw is 82 mA per board (fuse derated hold 0.22 A at 50 °C); power-rail current reaches 9.15 A (supply limit 10 A).

6.6 Signal-supply protection

Reverse polarity: the Schottky holds the DP50IP to at most -0.4 V until the fuse trips. Over-voltage: with the TVS alone, a supply set to $17\text{--}18\text{ V}$ holds the DP50IP at $16.7\text{--}16.9\text{ V}$ for seconds (above its 16.5 V maximum). A TL431/SCR crowbar was designed and simulated (trip 16.0 V , rail held below 16.06 V) but not adopted; the ES015's fixed 18 V setting limit and careful supply setting are relied on instead.

TO SUPPLY: These protection runs used a 1 A supply limit. Confirm whether the reversed-plug case should be re-run with the full 10 A available (now the planned setting).

6.7 Earlier verification round

Before the final values, a first round (ideal supplies, $R23 = 2.2\text{ k}$, $R37 = 33\ \Omega$) covered sequencing (signal first then power: clean; power first: harmless -2.75 A pulse), faults (Q12 open is the most dangerous: tens of amps per bank, limited only by the supply), safe operating area (hard reversal: 47.6 W in one device for 0.31 ms), and thermal behaviour (41.5 W worst-case dissipation at $\pm 5.8\text{ A}$).

7 PCB layout

7.1 Stack-up and fabrication

Four layers, 1.6 mm , 2 oz copper on all layers (0.07 mm), prepregs 0.1 mm , core 1.1 mm , ENIG, tented vias, S1000-2M (Tg 170), $8/8\text{ mil}$ minimum track/space, 0.2 mm minimum drill (U2 PowerPAD vias).

Layer	Use
L1 F.Cu	Components; COIL_DRIVE, COIL_RETURN, drive buses and the predrive trio; signal traces over SGND
L2 In1.Cu	PGND plane (power side) and SGND plane (signal corner), split under the DP50IP body; R27 is the only bridge
L3 In2.Cu	Power+ / Power− pours; Signal+ pour over the signal corner
L4 B.Cu	COIL_DRIVE and COIL_RETURN pours; Signal− pour over the signal corner

Table 9: Layer assignment.

7.2 Rules

Two via sizes: $0.6/0.3\text{ mm}$ (signal, $\approx 1\text{ A}$) and $1.0/0.5\text{ mm}$ (power, $\approx 2\text{ A}$). Zones use thermal reliefs for through-hole pads only (0.5 mm gap; spokes 1.5 mm on power zones, 1.0 mm on ground and signal zones, 2.0 mm pad overrides on J2, J11, U1 pins 1–8, D1, D2 and the bulk capacitors), island removal always, clearance 0.4 mm on outer and 0.3 mm on inner layers. A custom rule enforces 1 mm between power and signal nets. Copper keepouts of 8 mm diameter surround all 18 mounting holes (spacer footprint).

7.3 Routing principles

- The predrive pair and VBE_TAP run as a trio (VBE_TAP in the middle, 0.25 mm gaps), on one layer, crossing the ground split once, at least 1 mm from power copper.
- Each drive bus is one trace along its row with short taps to the base resistors; each base resistor sits at its transistor's base pin.

Net class	Clearance	Track	Via	Nets
Power	0.40	3.00	1.0/0.5	COIL_*, Power $\pm$, PGND, PRI_LINK1-3
Emitter	0.40	1.50	1.0/0.5	10 emitter-to-ballast nets
Drive_Bus	0.25	1.00	0.6/0.3	NPN/PNP_DRIVE_BUS
Base	0.25	0.60	0.6/0.3	10 output base nets
Drive	0.25	0.50	0.6/0.3	predrive pair, VBE_TAP, Q12-Q14 bases, sink
Signal_Power	0.30	1.00	0.6/0.3	Signal $\pm$, SIG_P/N_IN
Analog_Precision	0.30	0.30	0.6/0.3	Sense_value, COMMAND, J1 input, U2 -IN
SGND	0.30	0.50	0.6/0.3	SGND
Default	0.25	0.25	0.6/0.3	LEDs, E/D, status

Table 10: Net classes (mm). Diff-pair width equals track width so that Power $\pm$ and Signal $\pm$, which KiCad detects as pairs, route at the class width.

- COIL_DRIVE and COIL_RETURN run stacked or side by side to keep the high-current loop small; COIL_RETURN is a 20 mm pour (about 3 m Ω).
- Current sense: Is runs from U1 pin 9 to R25 beside J13 (the line carries current, so its length does not matter); J13 taps R25's pads directly; R25's low end has the single via pair to SGND next to J13's ground pin; the return flows in the SGND plane under the Is trace.
- SGND stays on the signal side: the L2 split runs under the DP50IP body, and R27 sits on the split near the board edge, away from the coil-return entry at U1 pin 4.

7.4 Layout checks

The PCB was compared with the schematic part by part (values, footprints, DNP flags) and pin by pin (283 pins), and its copper was traced geometrically for opens and shorts. The netlist used for the final simulations was rebuilt from the PCB's own connectivity, with parasitics extracted from the copper: predrive 214/189 mm (≈ 64 nH), VBE_TAP 171 mm (≈ 51 nH), and COIL_DRIVE capacitance of 1.58 nF to PGND and 0.69/0.63 nF to Power $\pm$.

Final state. All connections are routed: the four PNP base links, U2 pin 4 with C23/C24 to Signal-, and the C21 ground via, which the geometric check had flagged, were completed. DRC reports no unconnected items and no clearance errors; the only remaining DRC message is the 0.212 mm drill of the U2 PowerPAD vias against a 0.3 mm board rule, accepted deliberately (the board is ordered with a 0.2 mm minimum hole).

8 Mechanical and thermal design

8.1 Heatsink sandwich

The thirteen heatsinked devices lie tab-up under a HeatsinkUSA 8.460" profile (0.300" base, 1" fins) cut to 177.80 mm, spanning PCB $x = 99.06$ –276.86 mm and centred on the board (37.4 mm overhang each side). A finned aluminium plate (300 $\times$ 140 mm, 2 mm base, 19 fins) supports the board from below, fins down. Fourteen M3 $\times$ 20 clamp screws pass up through the bottom plate, a conical spring washer, a 6 mm metal spacer and the PCB into blind M3 threads in the heatsink; four corner screws hold the board to the plate with nylon lock nuts. The board carries no load in bending: the heatsink's weight goes through the screws and spacers into the bottom plate.

Layer (bottom to top, at a clamp screw)	mm
Bottom plate base (3.4 mm through hole)	2.00
Conical spring washer DIN 6796 (loaded)	0.75
Spacer, metal, smooth bore	6.00
PCB	1.60
MJL21194 body + Sil-Pad 900S	5.05 + 0.23
Subtotal	15.63
M3 × 20 ⇒ thread engaged in heatsink	4.37 (thread 5.0 deep, drill 6.5; about 0.45 lost to the countersink)

Table 11: Clamp stack. Kapton covers the plate’s top face with openings at every screw.

The insulator under every tab is Bergquist Sil-Pad 900S (SP900S-0.009-AC-1212: 0.229 mm, adhesive one side, 1.6 W/m·K, 0.61/0.47/0.41 °C·in²/W at 50/100/200 psi, 5500 V ac breakdown), about 0.8–1 K/W across a TO-264 tab. An earlier caliper reading of 0.35–0.40 mm probably included the liner; the driver and Q12 spacer heights below do not depend on the pad thickness, because the same pad sits on every device. Drivers and Q12 reach the TO-264 height on silicone rubber spacers (no FR-4): Q13/Q14 (4.70 mm body) on 2 × 0.2 mm silicone (12 % squeeze, add a third layer if loose); Q12 (2.55 mm body) on 3 × 1.0 + 0.2 mm silicone (22 % squeeze). Clamp torque is **1.8 in·lb (0.2 N·m)**, window 1.3–2.2, giving roughly 100–150 psi on each Sil-Pad; 10 in·lb would strip the aluminium threads and risk cracking packages.

8.2 Heatsink machining

Fourteen M3 × 0.5-6H holes, 5.0 mm full thread, 2.5 mm drill 6.5 mm deep, 90° countersink to 3.4 mm. Coordinates from datum end X0 and datum edge Y0 are $X = x_{\text{PCB}} - 99.06$, $Y = y_{\text{PCB}} - 32.56$; rows at $Y = 80.49$ (PNP row), 111.29 (centre pair), 142.09 (NPN row). The pattern is not symmetric, so X0 must be the end over the coil-output (J2) end of the board and Y0 the edge over the PNP row (KiCad $y = 70$ side).

8.3 Thermal

Worst continuous dissipation is about 43 W at ±5.7 A. With the heatsink at roughly 0.9–1.0 K/W in natural convection the heatsink reaches about 65–70 °C and the hottest junction about 80–85 °C (limit 150 °C). A fan along the fins would bring the heatsink to about 45 °C.

TO SUPPLY: Measured heatsink temperature at 7 A after 30 minutes. Put black tape on the aluminium first: bare aluminium has an emissivity of 0.03–0.1 and an IR thermometer reads it far too cold.

9 Bill of materials and ordering

The PCBWay BOM has 43 lines (105 placed parts per board), every line marked “no substitution”. The hand-fitted parts (Q2–Q14, U1; 14 per board) are listed separately for purchase and loose shipment. R27 is a copper net tie and is not a part. Board options: as in the stack-up section; remark “4L, 1.6 mm, 2 oz on all four layers”.

TO SUPPLY: Order reference, date, board quantity, and any parts PCBWay could not source.

10 Bring-up procedure

1. Before fitting the output devices: measure all ten TO-264 body thicknesses (within 0.15 mm ideally), bend leads in the jig, dry-fit, and check RV2's direction with an ohmmeter (pin 1 to wiper).
2. Set RV2 to the 5k end (zero bias). Set both supplies to 15.0 V with a 1.5 A current limit per rail for the first power-up.
3. Apply the signal supply first, then the power supply. Check the green LEDs and that both red LEDs are dark once the DP50IP has started.
4. Verify about 0.6 V across R37 (sink running) and that the spreader voltage moves with RV2.
5. Turn RV2 toward lower resistance until each bank carries 140 mA (trim with the heatsink warm).
6. Sign check before closing the loop: with the SIM960 in manual mode, a positive command must give a positive reading at J13. Closing the loop with the wrong sign is a runaway.
7. Raise the current limits, close the SIM960 loop with $P = 30$, $I = 2400 \text{ s}^{-1}$, D off (lower P toward 10 if the field-noise budget requires it, Section 1.1), and set the SIM960 output limits to about +9.3 V / −7.3 V so that loss of the sensor holds the coil near $\pm 7.6 \text{ A}$ instead of the stage maximum.
8. Calibrate the 70 ppm scale offset from the Measure-input loading against an external reference.
9. After the first warm-up, re-torque the clamp screws to 1.8 in·lb and compare tab temperatures across all ten devices with the thermal camera (black tape on every measured spot).
10. Measure the coil pair's resistance cold and after reaching equilibrium at 7 A to establish the temperature rise, and record the heatsink temperature in the same session.

11 Open items and residual risks

- Absolute bias depends on transistor models that disagree; trim on the bench and keep the 1.5 A limit for the first power-up.
- +7 A requires the coils to stay within about 29 K of ambient (about 0.42 K/W at 69 W).
- The field-noise requirement (1 mG) cannot be converted to a current specification until the coil field constant is measured; the DP50IP's own noise may dominate if it is near 10 G/A.
- No closed-loop noise or drift has been measured on any board yet.
- The SMBJ14A operates above its stand-off at 15 V; supply settings above 15.0 V are not allowed, and a supply set to 16–18 V is not fully caught by the TVS.
- The power connector J11 has no reverse-polarity protection.
- Fault cases with the final values (sensor lost, coil open/short in closed loop, device failures) were not re-run after the value changes.

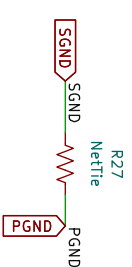
A Measurements still needed to complete this document

1. Coil field constant k_{coil} (G/A at the atoms), and the bandwidth and pk-pk/rms basis of the 1 mG requirement.
2. Coil temperature rise at 7 A (cold/hot resistance) and the coil pair's self-resonant frequency.
3. 1V1: RV2 pin 1–2 resistance (unpowered) at the 150 mA/bank trim; the three continuity readings on the suspect NPN.
4. Heatsink temperature at 7 A (black tape on the aluminium).
5. Closed-loop current noise once the loop is closed on the real coil.
6. PCBWay order reference, date and any parts that could not be sourced.

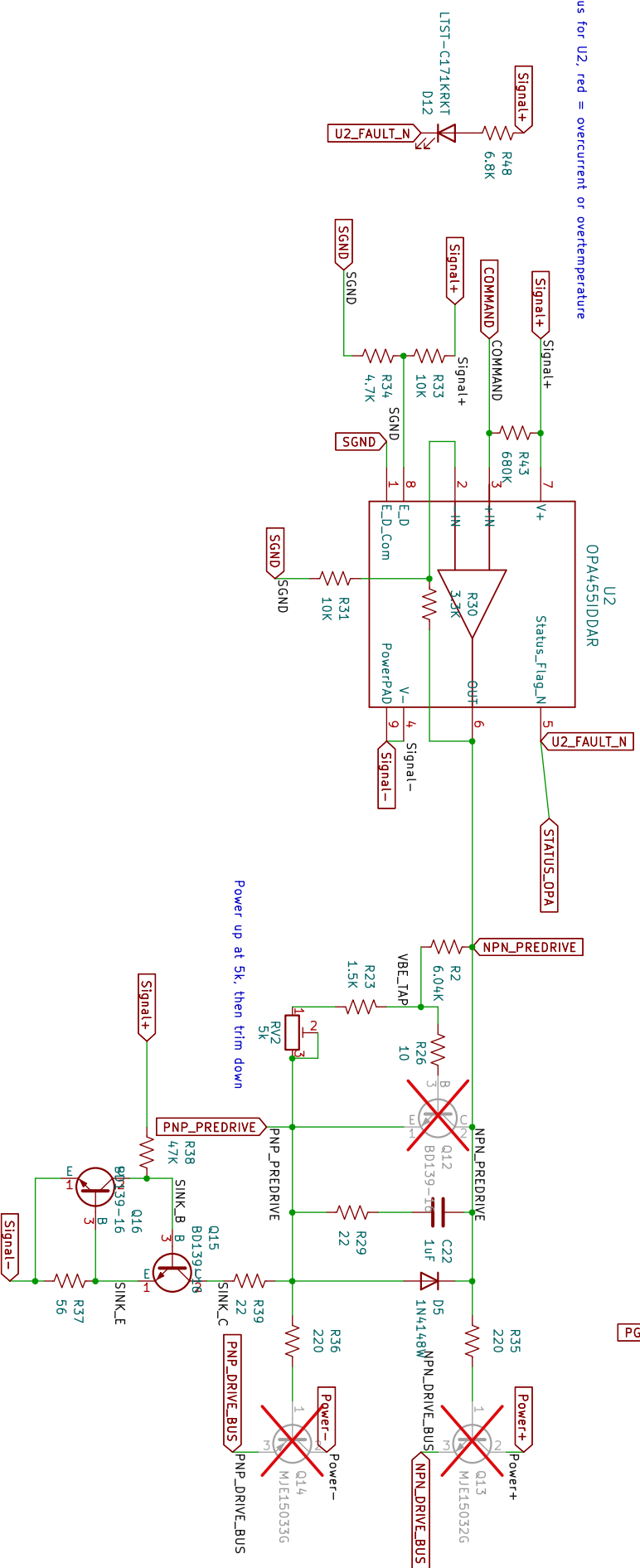
7. SIM960 P/I values chosen on the real system.
8. Photographs of the assembled board, the heatsink sandwich and the rack.

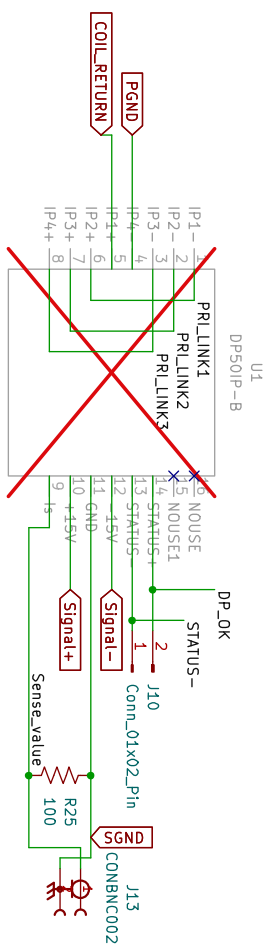
B Schematic

Star point SGND and PGND link. Only link the two grounds at U1 pin4 PGND

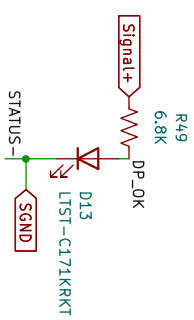


Status for U2, red = overcurrent or overtemperature

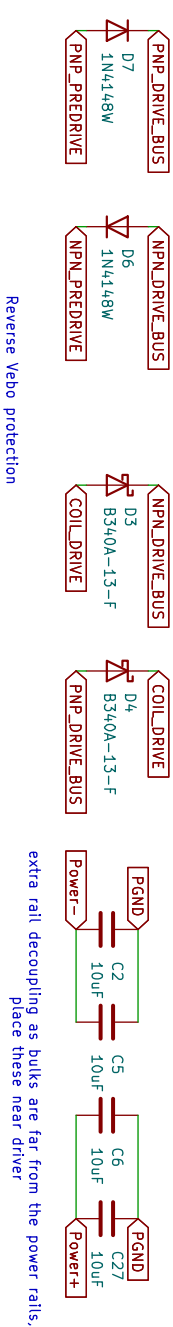
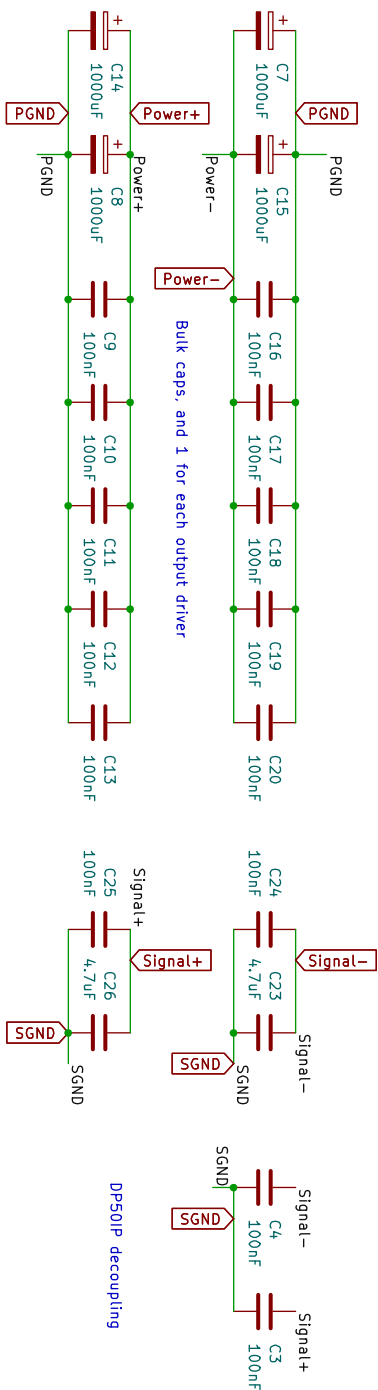




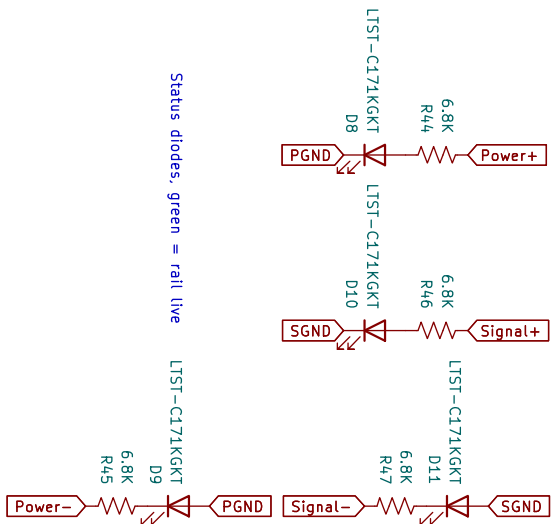
Status for DP, red = DP50IP not ok

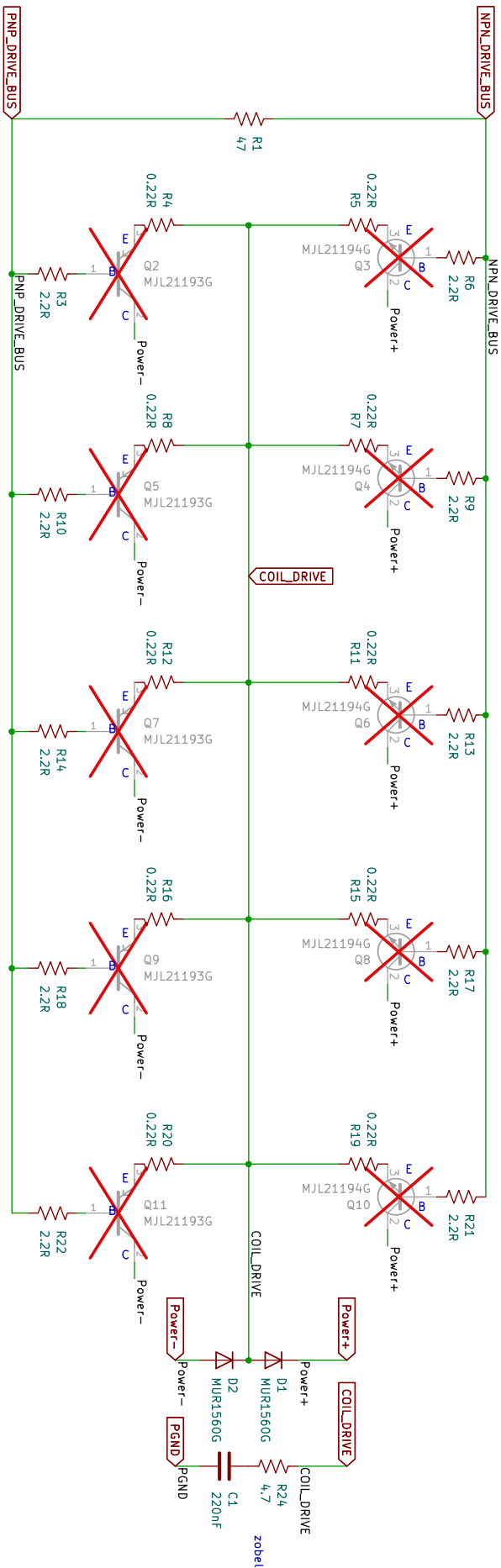


Op-amp decoupling



Reverse Vbeo protection





sim found possible oscillation, so add in case

